Combining Filling and Fusion Strategies for Generating Synthetic Daily Landsat Time Series Image on Google Earth Engine

Jiali Li, Xiaoping Liu, Yunfei Li, and Mengwei Liu

***Abstract*— Landsat satellites have provided high-quality Earth observations for more than 40 years, which significantly benefits much research on agriculture, environment, ecology, and so on. However, its low temporal resolution (16-day) and disturbances such as cloud contamination, prevent its usage in some scenarios. Therefore, reconstructing the Landsat image series is always an important topic. Currently, the approaches for reconstructing that are massive, which mainly include the filling-based methods and fusion-based ones. However, their scalability and applicability in large-scale applications are limited. To address this problem, based on the Google Earth engine (GEE), which is a powerful cloud platform, this article introduces a GEE-based fusion and filling model (GFFM) for generating high-quality synthetic Landsat surface reflectance time series data. This model adopts a pixel-wise regression technique to fuse the Landsat and Moderate Resolution Imaging Spectroradiometer (MODIS) data, providing a synthetic image series first. Then, a harmonic analysis is used to densify the Landsat image series. Finally, we utilize a Bayesian model average (BMA) as a weight function to integrate and adjust the previously obtained data to acquire the final seamless image series. We compare the proposed GFFM with some state-of-the-art fusion and filling approaches on various datasets. The experimental results demonstrate that the GFFM not only outperforms these fusion and filling approaches on different datasets, but also shows more robustness in cases of less and cloudy input data.**

***Index Terms*— Google Earth engine (GEE), Moderate Resolution Imaging Spectroradiometer (MODIS), spatiotemporal fusion, synthetic Landsat data.**

I. Introduction

HIGH-RESOLUTION remote sensing time series images are crucial for a comprehensive understanding of Earth's systems. Their importance is evident in various applications, such as forest detection [1], land cover dynamics monitoring [2], [3], urbanization tracking [4], and biomass estimation [5]. The widespread availability of shared remote sensing data, combined with advancements in high-performance cloud computing, has significantly eased the challenges in analyzing large-scale, long-term observational data for global change research. However, limitations in current technology and cost considerations often necessitate a tradeoff between spatial and temporal resolution in single satellite sensors. The Landsat provides precise radiometric calibration and high spatial resolution images from 1972, granting widespread use in Earth observation [6], [7], [8]. Nevertheless, its low temporal resolution, characterized by a 16-day revisit cycle, and disturbances such as cloud contamination and shadows can prevent the acquisition of clear images in cloudy regions for extended periods [9]. Reconstructing the Landsat image series is always an important topic in the remote sensing community.

To address the temporal resolution constraints of Landsat imagery and produce high-quality time series data, two main types of methods have been developed [10]. The first one is based on a filling strategy, primarily utilizing cloud-free high-resolution image series to address missing pixels. Filling methods can be categorized into temporal-based, spatial-based, and hybrid methods. Temporal-based methods, such as thresholding, harmonic analysis [11], [12], Savitzky–Golay (SG) [13], moving averages [14], fill missing pixels in the target image using one or more pixels from temporally adjacent high-quality images. For example, the harmonic analysis of time series (HANTS) method has been widely used for time-series data reconstruction [11], [15], [16], which uses a series of sinusoidal functions to decompose time series data. In contrast, spatial-based methods use data from surrounding pixels to interpolate and fill missing data in the target pixels [17]. Hybrid methods combine them, employing spatial and temporal data to rectify polluted images. For example, the neighborhood similar pixel interpolator (NSPI) method, a well-known hybrid method, is effective in eliminating image pollution pixels [18] and has been successfully utilized for stripe filling in Landsat 7 ETM+ images [19].

The second way to generate high-quality Landsat series data is fusion, which blends Landsat images and those having high temporal but low spatial resolution (such as Moderate-Resolution Imaging Spectroradiometer, MODIS) images. This type of fusion is called spatiotemporal fusion. Currently, fusion methods are instrumental in monitoring land dynamics [20], [21] and studying phenology [1], [5], [22]. Spatiotemporal fusion methods are generally categorized into five types according to their underlying principles:

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unmixing-based methods, weight function-based methods, Bayesian-based methods, learning-based methods, and hybrid methods [23]. The first fusion method may be the multisensor multiresolution technique (MMT) [24], an unmixing-based method, that integrates satellite images of different spatial and temporal resolutions. Based on the MMT, many unmixing-based fusion methods have been developed, they use linear spectral mixing theory to estimate the changes of fine pixels by analyzing and unmixing the composition of coarse pixels [2], [25]. The spatial and temporal adaptive reflectance fusion model (STARFM), notable as the most exemplary weight function-based method, predicts missing images via establishing the weight functions between pixels and their nearby pixels [26]. Subsequently, many methods have been derived from STARFM [27], [28]. Bayesian-based methods treat fusion as an issue of maximizing posteriori probability. Li et al. [29] utilized the Bayesian maximum entropy (BME) method to address multiscale challenges and capture complex spatial structures. In Bayesian-fusion, Xue et al. [30] advocated for integrating information from the image time series, notably enhancing the fusion results in heterogeneous landscape scenarios. With the advancement of machine learning, learning-based methods have gained prominence in fusion approaches. Examples include sparse representation [31], wavelet-artificial intelligence [32], extreme learning machine [33], and deep neural networks. Recent advances in deep learning such as convolutional neural networks (CNNs) [34], generative adversarial networks (GANs) [35], and transformer [36] have demonstrated superior performance in image fusion tasks by effectively capturing spatial and temporal dependencies across different datasets. To combine the advantages of various methods, hybrid methods have been developed. The flexible spatiotemporal data fusion (FSDAF) method merges concepts from unmixing, weight, and spatial interpolation, offering effective fusion in challenging scenarios [37]. The blend spatiotemporal temperatures (BLESTs) method integrates weight functions, temporal interpolation, and spatial downscaling to retain more spatial details and temporal variation information [38]. The newly proposed reliable and adaptive spatiotemporal data fusion (RASDF) method enhances estimation accuracy and stability by incorporating reliability indicators and optimization strategies [39].

Despite the plenty of filling and fusion methods available, challenges persist in generating extensive data of surface reflectance with high spatiotemporal resolution over long time series. A primary concern is their scalability. Most existing methods adopt a segregated approach to handling data, model algorithms, and computational resources. This separation often leads to considerable time and labor investment [40]. Second, methods applied to large-scale tasks need to be simple yet effective. Most existing filling and fusion approaches involve complex spatial operations with high computational demands to attain precise predictions. However, this computational intensity may limit their practicality in large-scale applications, such as national or continental-scale projects. Therefore, it's crucial to develop a method that efficiently bridges data gaps in reconstruction tasks while maintaining simplicity and effectiveness.

The Google Earth engine (GEE), a cloud platform for processing the remote sensing big data, can address the first concern aforementioned. The GEE revolutionizes the way we access and process data by providing seamless access to decades of imagery without the limitations of downloading, organizing, storing, and processing this information. For instance, GEE provides access to Landsat images from 1972 onward [41], enabling us to visualize and quantify changes in a scalable area over time. The highly scalable temporal adaptive reflectance fusion model (HISTARFM), a GEE-based fusion method, successfully built monthly-scale surface reflectance data for the continental U.S. from 2009 to 2021 [42]. This fusion approach integrates data, algorithms, and computing power in a cloud-based environment, effectively showcasing GEE's outstanding performance in large-scale data reconstruction tasks. However, this method is just designed for generating monthly-scale series data, which cannot satisfy the applications that require weekly or daily data. The daily-scale synthesis method offers high temporal resolution, allowing it to precisely capture subtle changes on Earth's surface [43], [44]. For the second problem, just adopting a filling or fusion strategy may be insufficient for two reasons. On the one hand, for both filling and fusion approaches, there is a tradeoff between their simplicity and effectiveness, that is why many existing approaches are complicated to ensure their performance. Under this circumstance, combining simple filling and fusion operations may bring better results than just using one of them. Moreover, the applicability of the filling and fusion strategy in different scenarios are different. The filling approaches have been reported that their efficacy largely depends on the availability of cloud-free images [40], [45]. Shu et al. [10] contrasted filling and fusion methods comprehensively and pointed out that in addition to time intervals and spatial heterogeneity, the factor that affects the accuracy of the filling methods is the cloud-contaminated size, while the fusion methods are affected by the difference in coarse and fine resolution scales. Therefore, legitimately combining the two strategies has the potential to obtain a more robust method.

In this study, we introduce a GEE-based fusion and filling model (GFFM) for generating high-quality synthetic Landsat surface reflectance time series data. Our method innovatively integrates filling and fusion strategies, processing time series data to forecast changes in spatial detail. Specifically, a pixel-wise regression technique and harmonic analysis are, respectively, used to generate preliminary fusion results and seamless Landsat data, we then utilize a Bayesian model average (BMA) as a weight function to integrate and adjust the previously obtained data. It can effectively utilize data sources of different resolutions, balancing temporal information and spatial consistency, and mitigate the limitations inherent in various data sources, thereby elevating the quality of the final fusion outcomes. In addition, the newly proposed GFFM can provide Landsat series with various temporal scales, including daily, weekly, and monthly.

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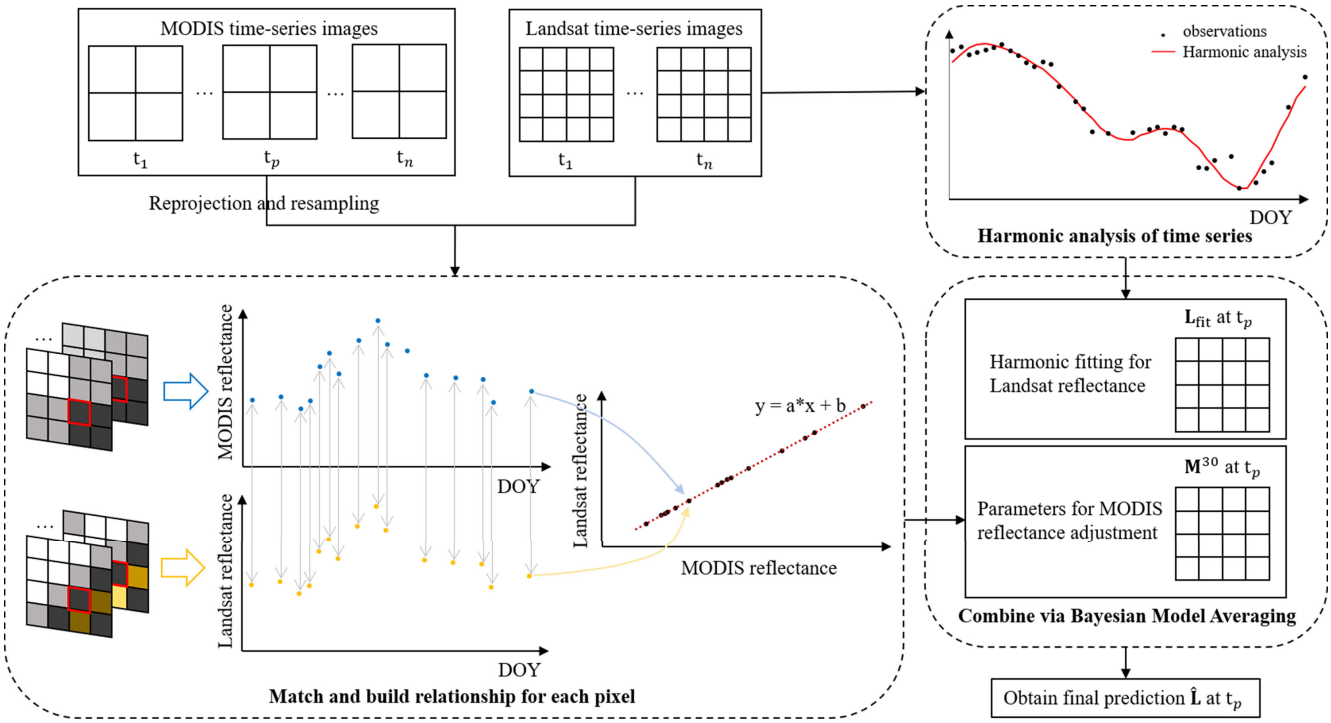


Fig. 1. Flowchart of the proposed GFFM.

The article is structured as follows. Section II details the GFFM method. Section III reports on experimental testing and results. A discussion is included in Section IV, and the conclusion is outlined in Section V.

II. Methodology

The proposed GFFM method is structured into three distinct phases, shown in Fig. 1. Initially, it fuses Landsat and MODIS data to obtain preliminary fusion results at a 30-m resolution. This step is followed by a filling strategy performing harmonic analysis on the Landsat time series data to enhance spatial detail. Finally, a weight function is utilized to integrate the previous fusion and filling results, resulting in the refined final output.

A. Fusion of Landsat and MODIS Images

The Landsat and MODIS image datasets are geographically calibrated to match their coordinate systems and resolutions, ensuring spatiotemporal congruence. To achieve this, MODIS images are rescaled to a 30-m resolution through bicubic interpolation, creating a pixel-by-pixel correspondence with Landsat images. This process guarantees consistency across different data sources, essential for accurate pixel-level analysis. Bicubic interpolation is a high-quality interpolation technique that takes into account information from adjacent pixels, thus preserving image detail [46]. Consequently, this technique yields enhanced image quality during downscaling from lower to higher resolutions. A highly efficient pixel-wise regression model is then developed to correlate the rescaled MODIS images and Landsat images within the time series framework. The regression coefficient θ is calculated using the ordinary least squares method for each pixel in each spectral band

$$\mathbf{M}_{i,b} \cdot \theta_{i,b} = \mathbf{L}_{i,b} \tag{1}$$

$$\theta_{i,b} = \left(\mathbf{M}_{i,b}^{T}\mathbf{M}_{i,b}\right)^{-1}\mathbf{M}_{i,b}^{T}\mathbf{L}_{i,b}. \tag{2}$$

Herein, the matrix $\mathbf{M}$ is defined as $[\mathbf{1}, \mathbf{m}_t]$, where $\mathbf{1}$ is a vector of ones, and $\mathbf{m}_t$ is a vector of clear and usable MODIS surface reflectance values within the specified time series. $\mathbf{L}$ represents the vector of Landsat surface reflectance values, corresponding in time to the MODIS data. The index i refers to the specific pixel being processed, and b indicates the spectral band. The vector θ consists of regression coefficients that encompass both the slope and the intercept of the regression model. This technique has been validated in previous research for its efficacy in correcting and spatially downscaling MODIS NDVI time series data [40], [47].

Utilizing the calculated regression coefficients, we establish initial fusion results that synergize information from both Landsat and MODIS. This is articulated as follows:

$$\widehat{M}_{i,b,t_p} = \mathbf{M}_{i,b,t_p} \cdot \theta_{i,b} \tag{3}$$

where $\widehat{M}$ represents the adjusted MODIS reflectance which has been integrated with the Landsat imagery and rescaled to a 30-m spatial resolution. $\mathbf{M}_{i,b,t_p}$ is expressed as $[1, m_{i,b,t_p}]$, where m_{i,b,t_p} denotes the surface reflectance value of pixel i at the predicted time t_p.

B. Harmonic Analysis of Landsat Series

To fill data gaps in the analysis of Landsat computed images, a harmonic (Fourier) model is meticulously constructed for each pixel and spectral band. Utilizing simple yet repeatable

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forms, this model efficiently deconstructs intricate curves into a series of simplified cosine waves, enriched with supplemented elements [48]

$$\tilde{\mathbf{L}}_{b,\mathbf{t}} = \beta_{0,b} + \sum_{i=1}^{N} A_{j,b} \cos\left(\omega_{j,b}\mathbf{t} - \phi_{j,b}\right) \tag{4}$$

$$\mathbf{L}_{b,\mathbf{t}} = \tilde{\mathbf{L}}_{b,\mathbf{t}} + \varepsilon_{b,\mathbf{t}} \tag{5}$$

where $\mathbf{L}$, $\tilde{\mathbf{L}}$, and ε, respectively, correspond to the original series, the reconstructed series, and the error series within Landsat computed imagery for spatiotemporal fusion. The spectral band is represented by b, while $\mathbf{t}$ represents the times of Landsat observations. The number of harmonics is denoted by N, where each harmonic j is characterized by its amplitude A, frequency ω, and phase ϕ. In addition, β_0 represents the average value of the observations series.

The formula expressed in (4) can be decomposed into individual cosine and sine components

$$\tilde{\mathbf{L}}_{\mathbf{t}} = \beta_0 + \sum_{j=1}^{N} \left[a_j \cos\left(\omega_j \mathbf{t}\right) + b_j \sin\left(\omega_j \mathbf{t}\right)\right] \tag{6}$$

where the coefficients a_j and b_j denote of triangular components for harmonic j within the suggested harmonic model.

The harmonic model is adaptable to account for periodicity by including higher frequency bands and their corresponding sine harmonic components. Models with higher frequency harmonics are more complex than lower frequency ones and offer superior modeling performance in addressing discrepancies in time series data. However, overfitting must be taken into account. As per previous studies, to ensure accuracy and stability in the time series model, the number of observations should be at least triple the number of estimated coefficients [6], [11]. Consequently, the selection of frequency value ω for harmonic analysis is based on the number of observations in the time series data. For instance, when there are between 9 and 15 distinct observations, a frequency value of 1 is used, which corresponds to three estimated coefficients. For 15–21 distinct observations, a frequency value of 2 is chosen, which corresponds to five estimated coefficients. In addition, when the observation range is 6–9, the model utilizes the lowest frequency 1, which corresponds to fitting only the trend of the time series using a limited set of coefficients, and for 1–5, the median of all the observations is used.

Parameters estimation is conducted through a systematic iterative fitting process. Initially, the harmonic model performs a least squares fit on all available data within a defined time span to produce a preliminary curve. Points of deviation are identified and removed by comparing this curve with the actual observational curve. This procedure continues iteratively until cloud contamination is effectively addressed, culminating in the precise determination of each estimated parameter.

C. Bayesian Model Averaging

To enhance the utilization of existing data and account for temporal relevance, BMA is employed to fully utilize preliminary fusion and filling results. This method combines the initial fusion results of high spatiotemporal resolution data acquired from the fusion of MODIS and Landsat with the filled outcomes from harmonic analysis, to refine the reconstruction results. BMA applies the Bayesian theorem to depict parameters within the model as specific probability distributions [49]. These probabilities serve as weights for a comprehensive weighted consideration to obtain a better model. Herein, outputs and variances from two information sources are used for weighted summation. This approach effectively combats offset distortion outcomes, reduces variances among models, and merges multiple data sources to achieve more accurate analytical results

$$\widehat{L}_{i,b,t_p} = \widehat{M}_{i,b,t_p} \frac{\rho_{t_p,\tilde{L}}}{\rho_{t_p,\hat{M}} + \rho_{t_p,\tilde{L}}} + \tilde{L}_{i,b,t_p} \frac{\rho_{t_p,\widehat{M}}}{\rho_{t_p,\widehat{M}} + \rho_{t_p,\tilde{L}}} \tag{7}$$

where $\widehat{L}$ represents the high-quality reconstructed result at the predicted time t_p. $\tilde{L}_{i,b,t_p}$ is the result of harmonic time series analysis fitting at t_p. The covariance between the harmonic analysis fitting result of Landsat data and the closest Landsat image at time t_p is denoted by $\rho_{\tilde{L}}$. Similarly, $\rho_{\widehat{M}}$ is the covariance between the downscaled MODIS surface reflectance result and the nearest Landsat image at time t_p.

When the interval between the forecast date and the observation dates of the nearby image exceeds ten days, and the RMSE of $\widehat{M}_{i,b,t_p}$ with the closet Landsat image at time t_p is three times higher than the RMSE of $\tilde{L}_{i,b,t_p}$ with the same Landsat image [6], it's supposed that the MODIS image has identified abrupt changes in surface features not captured by the adjacent Landsat image due to its lower temporal resolution. Consequently, the weight allocation of the adjustment results is modified. The formula for these revised results is outlined as follows:

$$\widehat{L'}_{i,b,t_p} = \widehat{M}_{i,b,t_p} \frac{\rho_{t_p,\widehat{M}}}{\rho_{t_p,\hat{M}} + \rho_{t_p,\tilde{L}}} + \tilde{L}_{i,b,t_p} \frac{\rho_{t_p,\tilde{L}}}{\rho_{t_p,\widehat{M}} + \rho_{t_p,\tilde{L}}} \tag{8}$$

where $\widehat{L'}$ represents the adjusted high-quality reconstructed result at the predicted time t_p.

III. Experiment

A. Data and Evaluation Metrics

Our evaluation of the GFFM method encompassed two widely recognized remote sensing image time series datasets. The first dataset is from the Coleambally Irrigation Area (CIA) in southern New South Wales, Australia. This area has been widely used in remote sensing spatiotemporal study, with notable agricultural phenological changes and high spatial heterogeneity. For the CIA, we selected 18 pairs of cloud-free Landsat-MODIS images from the year 2002. All Landsat imagery for the CIA dataset was sourced from the Landsat 7 ETM+ satellite. The false-color composites are depicted in Fig. 2. The secondary dataset is derived from the Lower Gwydir catchment (LGC) in northern New South Wales, Australia. We used 14 pairs of cloud-free Landsat-MODIS images from 2004. The Landsat imagery for the LGC dataset was obtained from the Landsat 5 TM satellite. As shown in Fig. 3, the LGC experienced a significant flood event in mid-December that led to sudden changes in land covers, making it a highly temporally dynamic research area. All data

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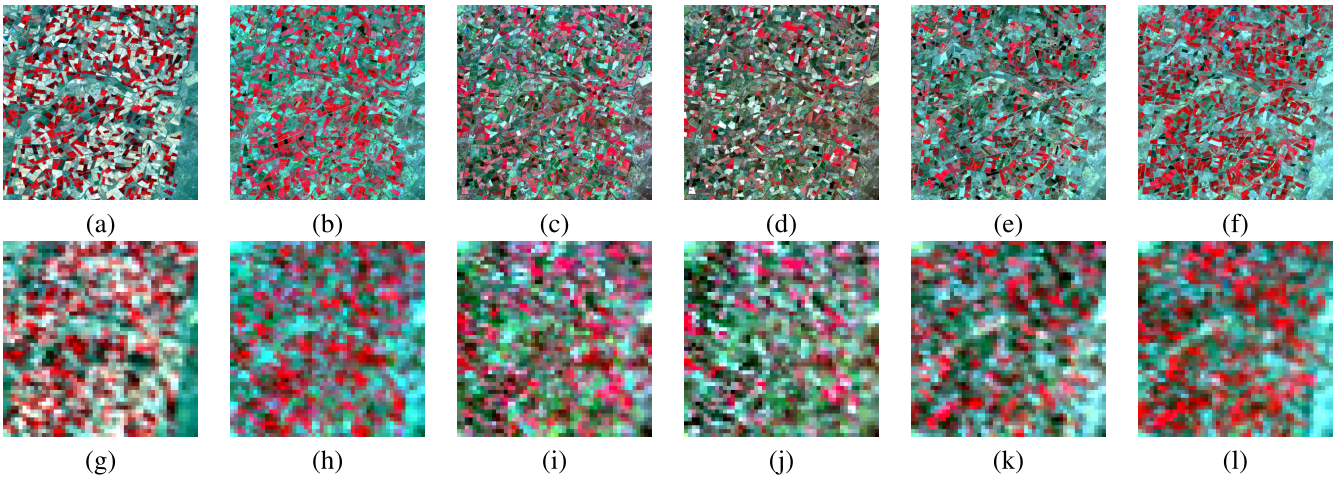


Fig. 2. Partial data of CIA. The image indexes of the original Landsat image corresponding to 2, 5, 9, 10, 13, and 16 are shown in (a)–(f), respectively. (g)–(l) Represent the corresponding MODIS images. The NIR, red, and green bands are used as RGB composition for all image demonstrations.

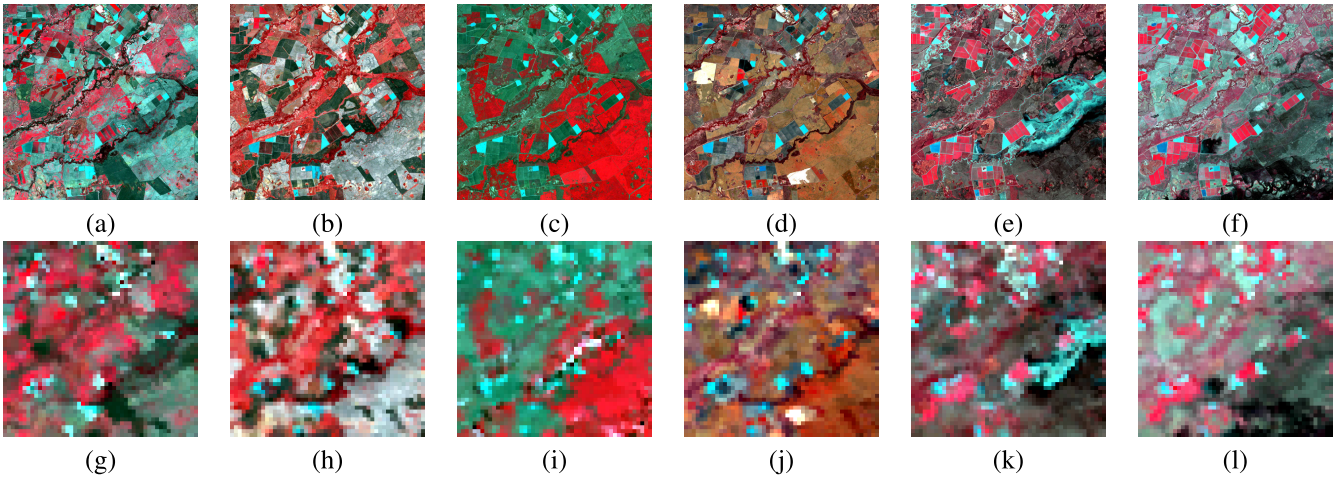


Fig. 3. Partial data of LGC. The image indexes of the original Landsat image corresponding to 2, 5, 8, 11, 13, and 14 are shown in (a)–(f), respectively. (g)–(l) Represent the corresponding MODIS images.

used in the experiment were retrieved from the GEE platform. The Landsat surface reflectance product was created using the Landsat Ecosystem Disturbance Adaptive Processing System (LEDAPS) algorithm [50]. The referenced MODIS surface reflectance data product is MOD09GA. The acquisition dates of all the images at the two study sites are displayed in Table I.

Our experiment employed root mean square error (RMSE), structural similarity (SSIM), correlation coefficient (CC), Erreur relative global adimensionnelle de synthese (ERGAS), and spectral angle mapper (SAM) as evaluation metrics for model performance. RMSE offers a comprehensive depiction of radiative discrepancies between fusion outcomes and actual observation images by directly comparing pixel value alterations. SSIM is a robust standard for assessing image quality, which assumes that human eye evaluation of image quality is based on structural information in the image. Therefore, good image quality should maintain the structural information in the restored image as much as possible to minimize distortion. To evaluate spatial detail similarity between fused and reference images, we utilized a high-pass filter to extract high-frequency details from reference images and computed CC values. ERGAS offers a comprehensive evaluation of the

TABLE I
ACQUISITION DATES OF CLOUD-FREE LANDSAT-MODIS IMAGE PAIRS USED IN EXPERIMENTS

Image index	Site1: CIA		Site2: LGC	
	DOY	Date	DOY	Date
1	4	04/01/2002	27	27/01/2004
2	11	11/01/2002	34	03/02/2004
3	43	12/02/2002	50	19/02/2004
4	52	21/02/2002	107	16/04/2004
5	68	09/03/2002	123	02/05/2004
6	75	16/03/2002	178	26/06/2004
7	91	01/04/2002	187	05/07/2004
8	100	10/04/2002	219	06/08/2004
9	107	17/04/2002	235	22/08/2004
10	123	03/05/2002	258	14/09/2004
11	219	07/08/2002	299	25/10/2004
12	228	16/08/2002	331	26/11/2004
13	235	23/08/2002	347	12/12/2004
14	244	01/09/2002	363	28/12/2004
15	267	24/09/2002		
16	283	10/10/2002		
17	292	19/10/2002		
18	331	27/11/2002		

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TABLE II
ASSESSMENT METRICS OF RECONSTRUCTION RESULTS ON THE CIA DATASET. VALUES WERE AVERAGED FOR ALL 16 RESULTS AND THE BEST VALUE IS HIGHLIGHTED IN BOLD

Band	GFFM			VIPSTF			ROBOT			ESTARFM			FSDAF			RASDF		
	RMSE	SSIM	CC	RMSE	SSIM	CC	RMSE	SSIM	CC	RMSE	SSIM	CC	RMSE	SSIM	CC	RMSE	SSIM	CC
Ideal	0	1	1	0	1	1	0	1	1	0	1	1	0	1	1	0	1	1
Blue	**0.0097**	**0.9522**	**0.8478**	0.0117	0.9387	0.8122	0.0099	0.9500	0.8406	0.0110	0.9445	0.8335	0.0120	0.9316	0.7905	0.0120	0.9308	0.7903
Green	**0.0115**	**0.9349**	**0.8457**	0.0131	0.9211	0.8109	0.0122	0.9286	0.8307	0.0126	0.9295	0.8326	0.0142	0.9082	0.7811	0.0143	0.9073	0.7814
Red	**0.0165**	**0.9222**	**0.8922**	0.0172	0.9200	0.8870	0.0176	0.9138	0.8812	0.0172	0.9193	0.8897	0.0199	0.8910	0.8428	0.0199	0.8912	0.8437
Nir	**0.0278**	0.9287	**0.9265**	0.0281	**0.9295**	0.9243	0.0290	0.9153	0.9134	0.0290	0.9248	0.9184	0.0346	0.8776	0.8669	0.0344	0.8797	0.8682
SWIR1	0.0352	**0.8872**	**0.8793**	0.0377	0.8768	0.8655	**0.0350**	0.8858	0.8783	0.0369	0.8847	0.8740	0.0415	0.8518	0.8375	0.0415	0.8525	0.8381
SWIR2	0.0295	**0.9153**	**0.9089**	0.0313	0.9076	0.8990	**0.0292**	0.9144	0.9087	0.0306	0.9116	0.9032	0.0347	0.8819	0.8696	0.0347	0.8834	0.8709
	ERGAS	SAM		ERGAS	SAM		ERGAS	SAM		ERGAS	SAM		ERGAS	SAM		ERGAS	SAM	
All	**0.7984**	**0.0632**		0.8699	0.0638		0.8699	0.0681		0.8437	0.0689		0.9587	0.0801		0.9612	0.0777	

overall spectral and spatial accuracy of the predicted image compared to the original reference. SAM facilitates the assessment of spectral similarities between the predicted result and reference data. A smaller RMSE, ERGAS, and SAM denote better predictive performance, while a bigger CC and SSIM indicate better-predicted performance. The integration of these evaluation metrics allowed for a comprehensive understanding of the proposed model's performance and suitability.

B. Prediction of Time Series Images on Heterogenous Area (Site 1)

To test the performance of the GFFM algorithm, we used 18 pairs of images from the CIA dataset, with 16 pairs in between dates designated for prediction. For each prediction image, all other images apart from the Landsat image of the predicted data were used as input data for predicting fusion outcomes. We compared our method with five other algorithms: the flexible virtual image pair-based spatiotemporal fusion (VIPSTF) approach [51], the robust optimization-based fusion (ROBOT) model [52], the enhanced SATRFM [27], FSDAF, and RASDF. ESTARFM is a weight function-based method, while FSDAF, VIPSTF, and RASDF are hybrid methods. The newly proposed ROBOT is a brand-new optimization-based method. Default parameters were used for comparative methods, which can be found in their corresponding articles. The VIPSTF and ROBOT algorithms had identical data inputs to GFFM. The ESTARFM algorithm used two image pairs nearest to the prediction date as input. Meanwhile, FSDAF and RASDF used image pair data from a date prior to the prediction date as input.

A comparison of the accuracy assessments of the six reconstruction methods in Fig. 4 reveals that the GFFM method achieved the most accurate predictions on most forecast dates. It is observed that methods using multiple image pairs as input data, such as GFFM, ROBOT, VIPSTF, and ESTARFM, generally outperform those using a single image pair. By comparing the scatter plots of the fusion outcomes shown in Fig. 5 it is clear that the GFFM method has superior aggregation effects at the 1:1 line between the predicted results and the actual values compared to the other five methods. Fig. 6 displays the visual comparison results of different reconstruction methods, where the third row shows the mean absolute error (MAE) between the predicted results of the subarea and the actual values. Notably, Fig. 6(a) means the MAE between the image at t_p and image at t_{p-1}. The figures clearly show that the predicted results of GFFM are most akin to the actual images in overall effectiveness, with the subarea displaying more abundant detail effects and the successful capture of minor features. The visualized results reveal a superior performance with smaller errors. Table II shows that the GFFM method achieves the best values on the most bands across all prediction results for each indicator. Specifically, the optimal RMSE, SSIM, and CC values, respectively, denote minor radiative differences, less spatial distortion, and a robust correlation with the actual images across the bands. A lower ERGAS value indicates that the predicted result is closer to the reference image. Smaller SAM shows better spectral similarity.

C. Experiment on Land Cover Change Area (Site 2)

The performance of the GFFM was tested using 14 pairs of images from the LGC dataset, with image data from December 12, 2004 of abrupt changes selected as the prediction image. The remaining images, excluding the Landsat image of the predicted date, were used as input data to predict reconstruction results. Five other algorithms: VIPSTF, ROBOT, ESTARFM, FSDAF, and FSDAF 2.0 were selected as comparison methods. The data input for VIPSTF and ROBOT algorithms was the same as for the GFFM algorithm. The ESTARFM algorithm used the two image pairs closest to the predicted date as input, while FSDAF and FSDAF 2.0 used the image pair data before the predicted date as input.

As depicted in Figs. 7 and 8, the visual comparison results of different reconstruction methods and the absolute discrepancy between the subregion prediction results and the actual values are presented. In the area of sudden land changes due to flash floods, subarea A with minor flood impact and subarea B with major flood impact were selected for detailed display. It can be observed that in subarea A, showed in the second row of Fig. 7 and the first row of Fig. 8, the method proposed shows the best fusion performance, followed by the ROBOT and ESTARFM methods. In subregion B, displayed in the third row of Fig. 7 and the second row of Fig. 8, a comparison of the outcomes of various algorithms indicates that GFFM and ROBOT closely resemble the actual image and provide better prediction results, whereas the VIPSTF method exhibits color distortion. Table III indicates that the GFFM method generally achieved optimal values in the blue, green, red, and SWIR1 bands. However, the NIR and SWIR2 band's optimal values were primarily obtained by the time series algorithms ROBOT and VIPSTF, with GFFM ranking as suboptimal. An analysis of the

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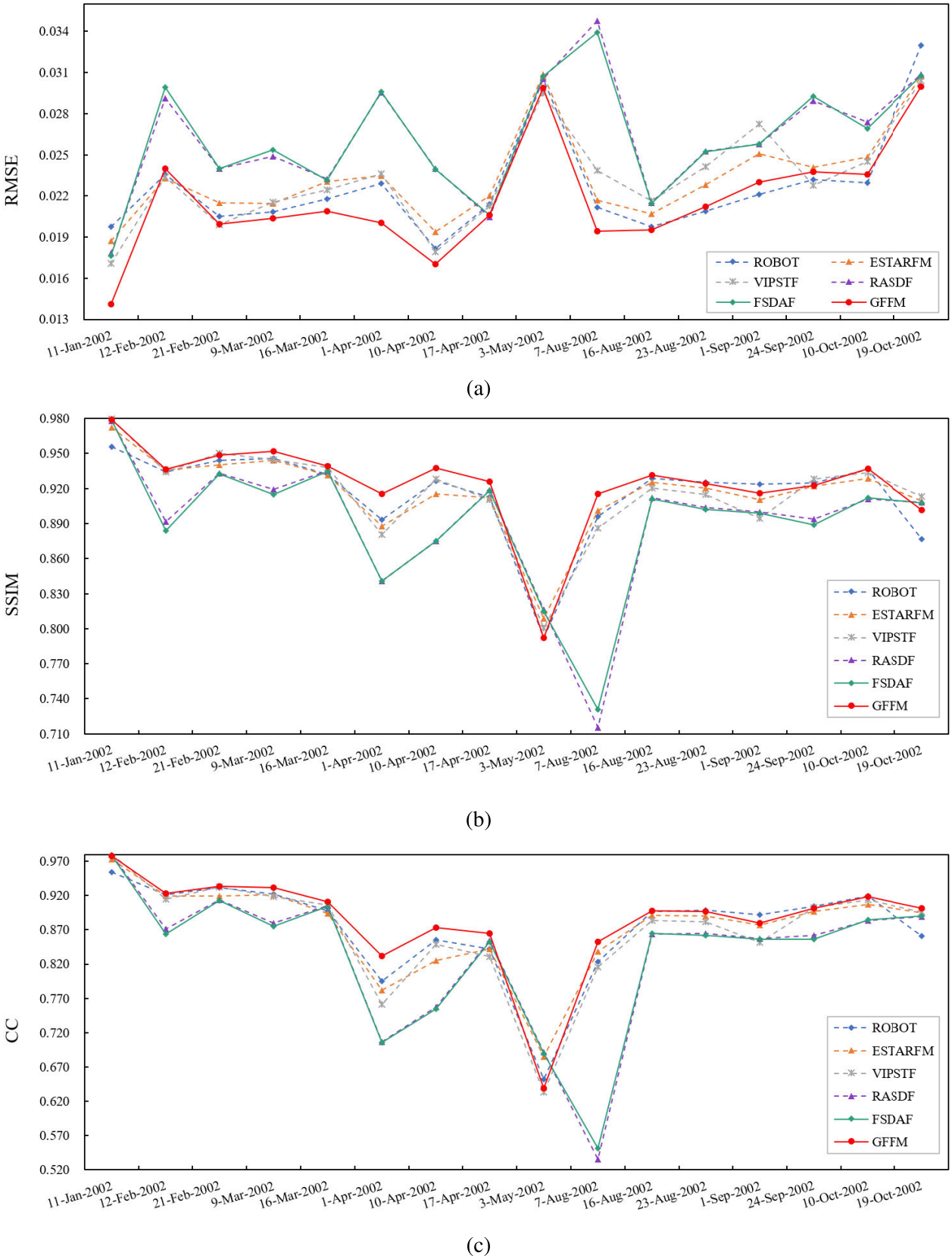


Fig. 4. Evaluation metrics of all the reconstruction results on the CIA dataset among six methods. Values were averaged over the six bands. (a) RMSE value. (b) SSIM value. (c) CC value.

computational principles of the comparison methods suggests that GFFM may be limited by pixel-level information and unable to fully capture the contextual information obtained by neighborhood or block computation methods. This limitation

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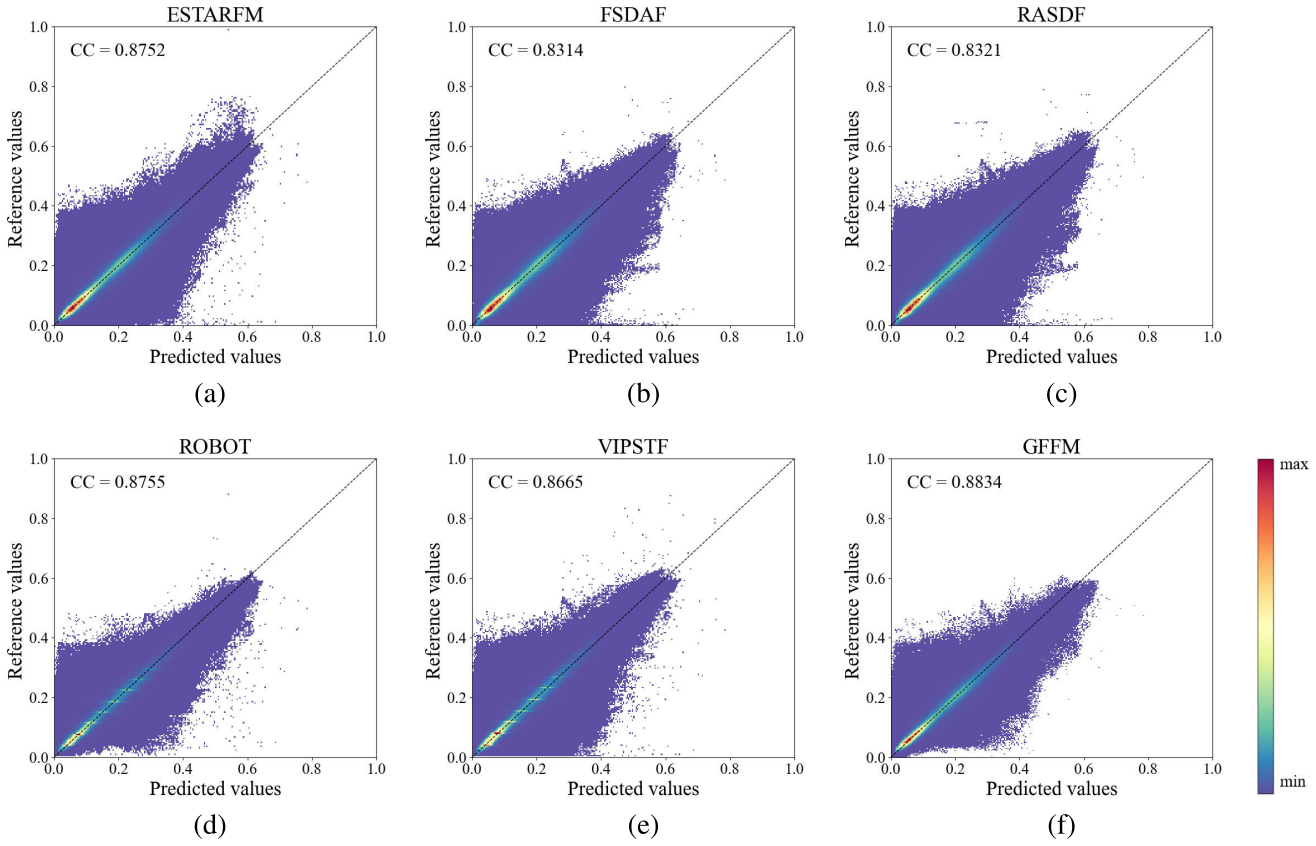


Fig. 5. Scatter plot of all pixels in the 16 reconstruction results on the CIA dataset. CC values were averaged across all the six bands. (a)–(f) Represent the ESTARFM, FSDAF, RASDF, ROBOT, VIPSTF, and GFFM predicted results, respectively.

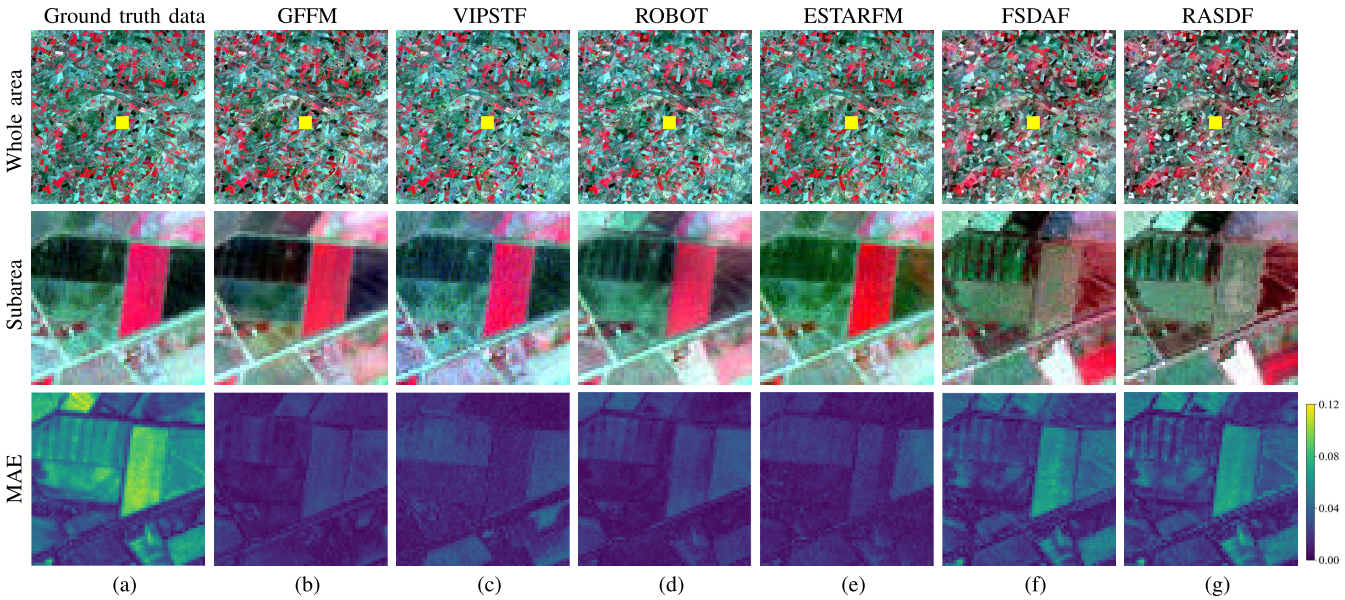


Fig. 6. Visual comparison of different predicted results on the CIA dataset. (a) Ground truth Landsat data was acquired on August 7, 2002. (b)–(g) Represent the GFFM, VIPSTF, ROBOT, ESTARFM, FSDAF, and RASDF predicted results, respectively.

results in relatively poorer performance in certain bands, particularly when dealing with spatially continuous features such as water bodies.

D. Comparison With GEE-Based Methods

The HISTARFM database, generated by the fusion of Landsat and MODIS time-series data, provides seamless surface reflectance time-series data at a 30-m spatial resolution. It currently allows for the indexing of monthly reflectance data from 2009 to 2021 for the mainland United States [42]. Zhou et al. [12] implemented the HANTS algorithm open source software on the GEE, which can scalably reconstruct surface observation series data. To comprehensively analyze the predicted performance of the method proposed in this article on the GEE platform, three zones with different surface covers were selected in the United States. GFFM was utilized

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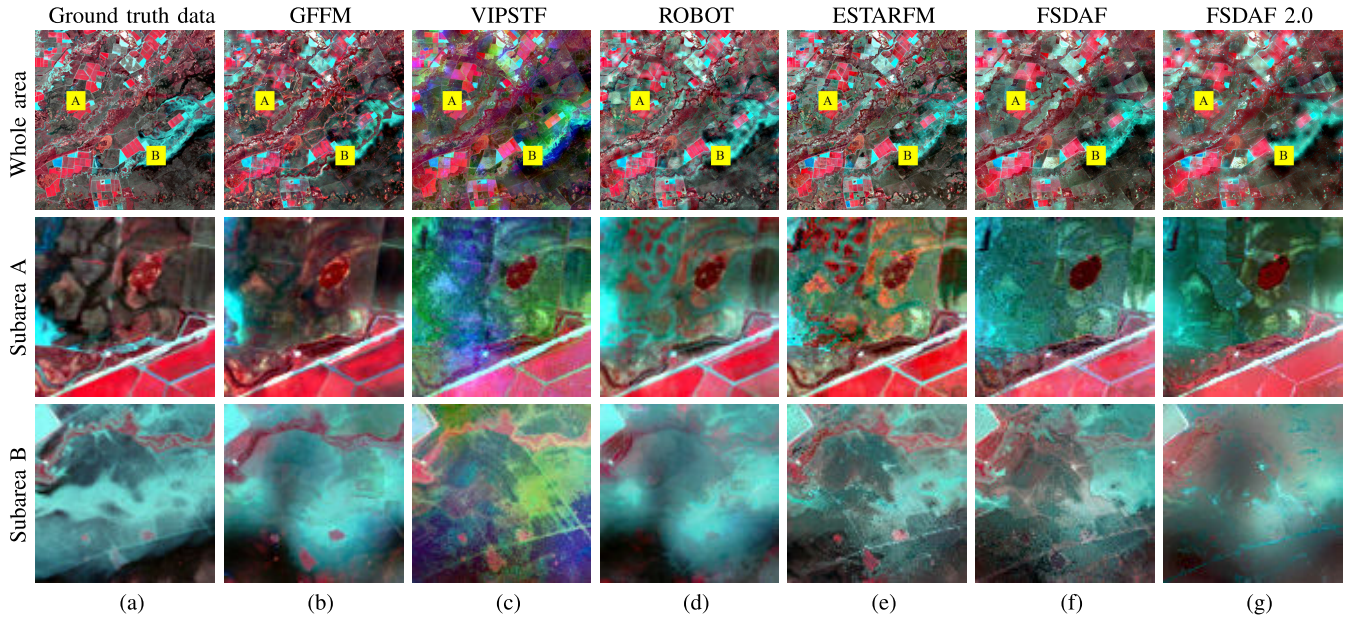


Fig. 7. Visual comparison of different predicted results on the LGC dataset. (a) Ground truth Landsat data was acquired on December 12, 2004. (b)–(g) GFFM, VIPSTF, ROBOT, ESTARFM, FSDAF, and FSDAF 2.0 predicted results, respectively.

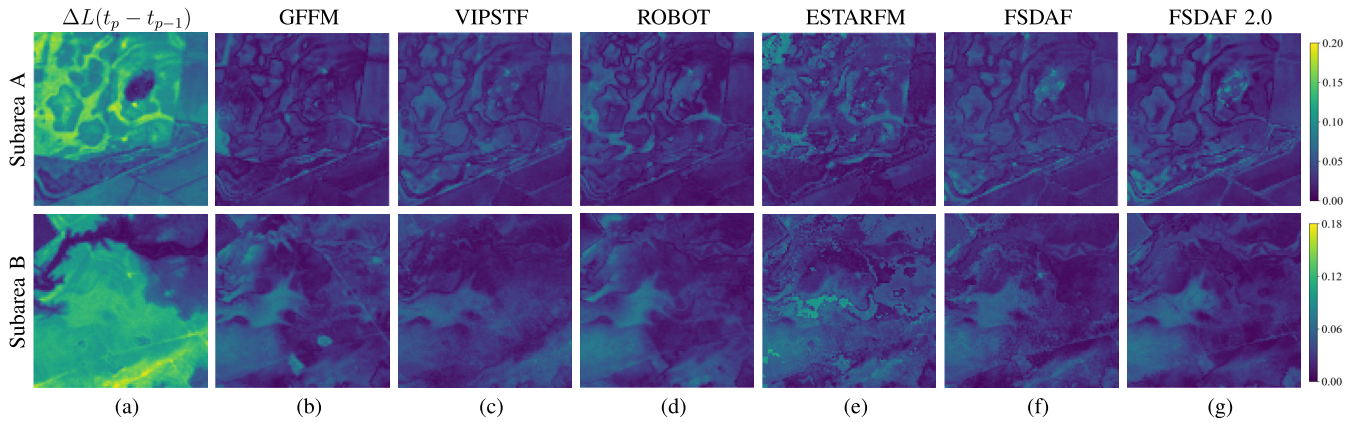


Fig. 8. MAE map on the subareas of the LGC dataset. (a) MAE between the image at t_p and the image at t_{p-1}. (b)–(g) Represent the MAE map of GFFM, VIPSTF, ROBOT, ESTARFM, FSDAF, and FSDAF 2.0, respectively.

TABLE III

METRICS OF EACH BAND OF ALL THE PREDICTION RESULTS ON THE LGC DATASET. THE OPTIMAL VALUES ARE PRESENTED IN BOLD

Band	GFFM			VIPSTF			ROBOT			ESTARFM			FSDAF			FSDAF 2.0		
	RMSE	SSIM	CC	RMSE	SSIM	CC	RMSE	SSIM	CC	RMSE	SSIM	CC	RMSE	SSIM	CC	RMSE	SSIM	CC
Ideal	0	1	1	0	1	1	0	1	1	0	1	1	0	1	1	0	1	1
Blue	**0.0156**	**0.8687**	**0.7274**	0.0190	0.7941	0.5491	0.0166	0.8397	0.6572	0.0165	0.8595	0.7056	0.0168	0.8502	0.6832	0.0176	0.8412	0.6647
Green	**0.0232**	**0.8173**	**0.7384**	0.0285	0.6932	0.5394	0.0238	0.7793	0.6728	0.0238	0.8040	0.7081	0.0238	0.7958	0.7004	0.0245	0.7888	0.6919
Red	**0.0276**	**0.7831**	**0.7159**	0.0291	0.7328	0.6932	0.0291	0.7466	0.6612	0.0289	0.7777	0.7005	0.0285	0.7620	0.6865	0.0291	0.7521	0.6767
Nir	0.0360	0.8491	0.8380	0.0367	**0.8561**	0.8471	**0.0350**	0.8430	0.8239	0.0351	0.8496	0.8327	0.0351	0.8511	**0.8499**	0.0384	0.8168	0.8139
SWIR1	**0.0548**	0.7332	**0.7533**	0.0588	0.6966	0.7300	0.0551	**0.7480**	0.7483	0.0689	0.6609	0.6427	0.0632	0.6868	0.7059	0.0666	0.6482	0.6694
SWIR2	0.0483	0.6771	0.6758	0.0564	0.5881	0.6381	**0.0452**	**0.7114**	**0.6833**	0.0592	0.6100	0.5628	0.0530	0.6576	0.6755	0.0540	0.6375	0.6729
	ERGAS	SAM		ERGAS	SAM		ERGAS	SAM		ERGAS	SAM		ERGAS	SAM		ERGAS	SAM	
All	1.4627	**0.1832**		1.5976	0.2034		**1.4275**	0.1898		1.6492	0.1992		1.5376	0.1911		1.6035	0.1843	

to calculate the monthly scale time-series data for 2018 and compared it with the HISTARFM and HANTS results. Zone 1 is a farmland region in Moses Lake, Washington State. Zone 2 is located in Brush Creek, Ohio, where the selected region is predominantly covered by deciduous forest vegetation. Zone 3 is situated in Washington, DC, USA, a highly heterogeneous urban area mainly composed of built-up areas and mixed vegetation. It's important to note that the input data for the method includes all available clean images and partially contaminated images. However, accuracy calculations and analyses of the predicted results are only conducted when the proportion of the actual usable data area on the surface exceeds 50%.

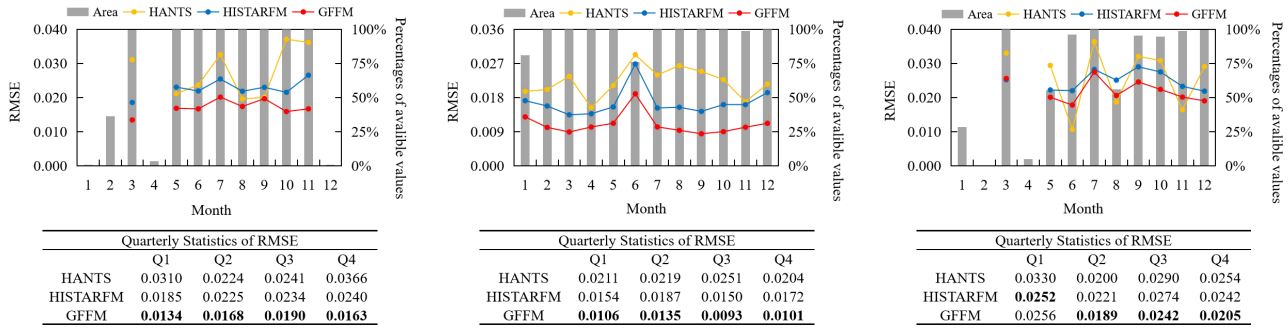
Table IV reflects the accuracy evaluation of the calculable time-series data in 2018 within the three areas, with the data in each area averaged over six bands. It can be observed that GFFM performs well on almost all indicators

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TABLE IV

ASSESSMENT METRICS OF PREDICTED RESULTS AMONG THREE U.S. ZONES. VALUES WERE AVERAGED FOR ALL CALCULATED RESULTS AND THE BEST VALUE IS HIGHLIGHTED IN BOLD

			Zone1			Zone2			Zone3		
	Band	Ideal	GFFM	HISTARFM	HANTS	GFFM	HISTARFM	HANTS	GFFM	HISTARFM	HANTS
RMSE	Blue	0	**0.0078**	0.0124	0.0126	**0.0086**	0.0135	0.0201	**0.0189**	0.0198	0.0257
	Green	0	**0.0114**	0.0141	0.0186	**0.0085**	0.0106	0.0204	0.0226	**0.0212**	0.0258
	Red	0	**0.0178**	0.0210	0.0260	**0.0101**	0.0278	0.0250	**0.0278**	0.0351	0.0321
	NIR	0	**0.0258**	0.0332	0.0466	**0.0165**	0.0120	0.0357	0.0249	**0.0246**	0.0386
	SWIR1	0	**0.0211**	0.0314	0.0361	**0.0122**	0.0206	0.0186	**0.0235**	0.0286	0.0389
	SWIR2	0	**0.0148**	0.0241	0.0260	**0.0095**	0.0146	0.0130	**0.0141**	0.0203	0.0285
	Mean	0	**0.0164**	0.0227	0.0276	**0.0109**	0.0165	0.0221	**0.0219**	0.0249	0.0263
SSIM	Blue	1	**0.9595**	0.9416	0.9209	**0.9633**	0.9265	0.7660	**0.8548**	0.8529	0.8360
	Green	1	**0.9482**	0.9437	0.8984	**0.9652**	0.9571	0.8306	0.8453	**0.8671**	0.8407
	Red	1	**0.9415**	0.9189	0.8959	**0.9652**	0.9490	0.8253	0.8461	**0.8533**	0.8468
	NIR	1	**0.9432**	0.9341	0.8536	**0.9535**	0.9264	0.9325	**0.9261**	0.8927	0.8599
	SWIR1	1	**0.9164**	0.8818	0.8634	**0.9660**	0.9350	0.9611	**0.9269**	0.8854	0.8299
	SWIR2	1	**0.9565**	0.9188	0.9125	0.9601	0.9449	**0.9633**	**0.9558**	0.9052	0.8650
	Mean	1	**0.9442**	0.9232	0.8908	**0.9622**	0.9398	0.8798	**0.8925**	0.8761	0.8464
CC	Blue	1	**0.9256**	0.9105	0.8802	**0.9033**	0.8657	0.6742	0.7704	**0.7836**	0.7755
	Green	1	0.9330	**0.9339**	0.8795	**0.9453**	0.9280	0.8245	0.7849	**0.8197**	0.8044
	Red	1	**0.9417**	0.9381	0.8915	**0.9528**	0.9327	0.8308	0.8038	**0.8171**	0.8094
	NIR	1	**0.9407**	0.9375	0.8513	**0.9566**	0.9338	0.9339	**0.9154**	0.8828	0.8323
	SWIR1	1	**0.9057**	0.8999	0.8590	**0.9658**	0.9338	0.9569	**0.9152**	0.8690	0.8651
	SWIR2	1	**0.9487**	0.9167	0.9094	**0.9550**	0.9323	0.9533	**0.9422**	0.8759	0.8501
	Mean	1	**0.9326**	0.9228	0.8785	**0.9465**	0.9210	0.8623	**0.8553**	0.8413	0.8228



Quarterly Statistics of RMSE

	Q1	Q2	Q3	Q4
HANTS	0.0310	0.0224	0.0241	0.0366
HISTARFM	0.0185	0.0225	0.0234	0.0240
GFFM	**0.0134**	**0.0168**	**0.0190**	**0.0163**

(a)

Quarterly Statistics of RMSE

	Q1	Q2	Q3	Q4
HANTS	0.0211	0.0219	0.0251	0.0204
HISTARFM	0.0154	0.0187	0.0150	0.0172
GFFM	**0.0106**	**0.0135**	**0.0093**	**0.0101**

(b)

Quarterly Statistics of RMSE

	Q1	Q2	Q3	Q4
HANTS	0.0330	0.0200	0.0290	0.0254
HISTARFM	**0.0252**	0.0221	0.0274	0.0242
GFFM	0.0256	**0.0189**	**0.0242**	**0.0205**

(c)

Fig. 9. Monthly and quarterly RMSE of predicted results in 2018 among three U.S. zones. Values were averaged over six bands. (a) Zone 1. (b) Zone 2. (c) Zone 3.

through three different land cover types. Fig. 9 displays the monthly and quarterly RMSE calculation of the time-series data in 2018 within the three areas, showing that the RMSE index of GFFM within each area outperforms HISTARFM. The fusion-filling algorithm GFFM and fusion method HISATRFM demonstrated superior stability compared to the filling algorithm HANTS. The reliability of temporal interpolation for high spatial resolution data largely depends on cloud-free image observations and relatively simple and constant temporal change patterns [40]. Consequently, interpolating surface reflectance with weak correlations can result in considerable variability in the outcomes. The predicted result data for July 2018, as shown in Fig. 10, reveals that the predictive results of the GFFM method are visually closest to the actual Landsat surface reflectance.

IV. DISCUSSION

A. Superiority of the GFFM Method

Most of the current fusion and filling methods require complex calculations within the spatial neighborhood, which can be time-consuming and labor-intensive when dealing with large-scale studies that necessitate efficient data processing [42], [53]. Moreover, these methods are not readily adaptable for parallel execution in contemporary cloud computing platforms (GEE). The method proposed in this article requires time-series data as input, which is a larger and more comprehensive data requirement compared to traditional fusion methods that typically only necessitate one or two pairs of images. GEE not only maintains the data but also provides significant computational power for processing, allowing users to access and quantify changes from dozens or even hundreds

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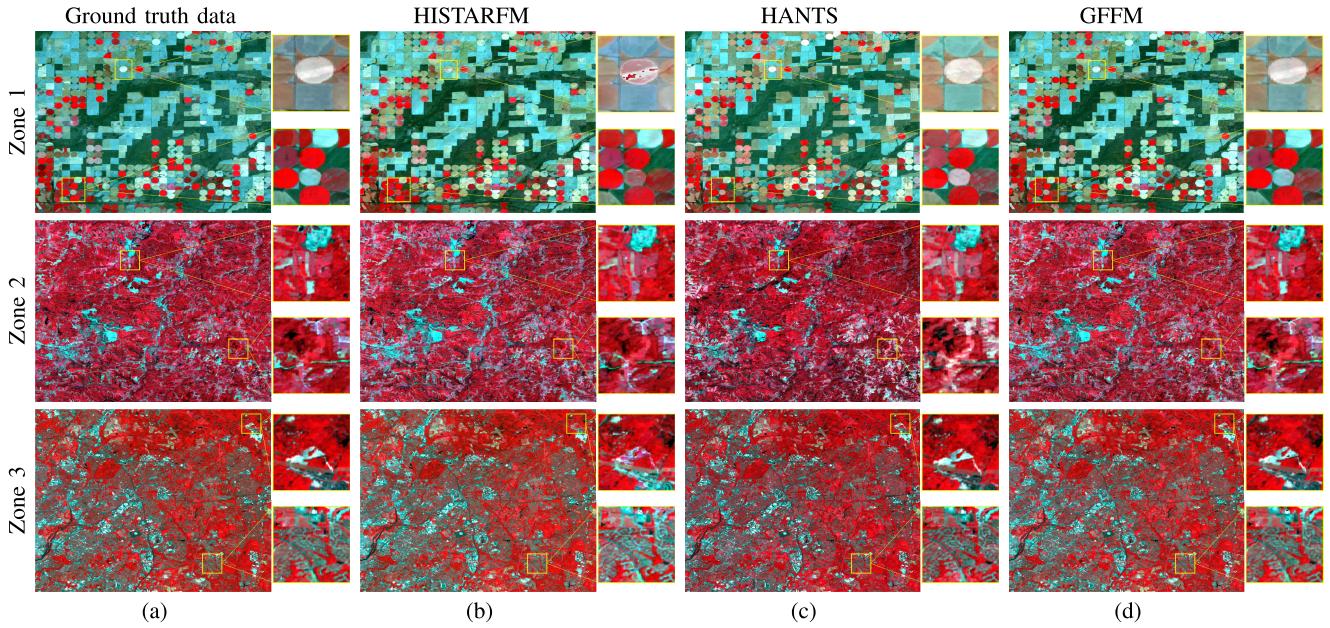


Fig. 10. Predicted results of different GEE-implemented methods on zone1 (farmland), zone2 (forest), and zone3 (urban). (a) Ground truth Landsat data in June 2018. (b) Estimated results from HISTARFM. (c) Estimated results from HANTS. (d) Estimated results from GFFM.

of time-series images over decades. By implementing image reconstruction in a cloud computing environment (GEE), the method not only benefits from the ease of data acquisition but also improves the accessibility of the image reconstruction dataset for further investigation. The method's superiority is demonstrated through its quantitative and qualitative comparisons with the predictive results of various state-of-the-art reconstruction algorithms in Section III.

In our specific research, our method excelled in three key aspects. First, it demonstrated its applicability at large spatial scales. By integrating with the GEE platform, we successfully achieved seamless cloud deployment of data, model algorithms, and computing power, thereby paving the way for efficient processing of large-scale data. Table V presents the overall running time and memory usage for four reconstruction methods to complete Section III-B, with all experiments conducted under the same computing conditions (i9-12900KF 3.2 GHz CPU and 16 G RAW). In Table V, GFFM and HANTS exhibit superior computational efficiency compared to VIPSTF and ESTARFM, primarily due to multitask parallel computing capabilities on the GEE platform and its ability to process data across any area. Notably, GFFM outperformed HANTS in terms of both processing speed and memory usage, making GFFM a practical method for large-scale reconstruction tasks, as it can handle extensive data without compromising accuracy or requiring excessive computational resources. This analysis shows that GFFM can effectively meet the demands of large-scale data management. Second, we adopted a unique approach that combines filling and fusion strategies. By fusing MODIS and Landsat data, we leveraged the benefits of both high temporal resolution and fine spatial resolution data. Furthermore, by considering temporal information and spatial consistency through harmonic analysis and the Bayesian average model, we significantly enhanced the overall consistency and accuracy of the data. Finally, our method effectively addressed the vacancy problem of the base image caused by continuous missing values, and successfully reconstructed high-resolution image time series data through pixel-based image synthesis methods.

The experimental results demonstrate that GFFM can effectively reconstruct data in diverse settings, including phenological change areas (e.g., CIA), flood disaster zones (e.g., LGC), and heterogeneous urban areas. By generating high-resolution, seamless surface reflectance data, GFFM offers robust support for various research and practical applications. For example, it can provide critical data for tracking dynamic environmental events such as flood disasters and urban heat islands, support precision agriculture tasks like crop health assessment and irrigation management, and facilitate urban planning while monitoring urbanization trends. These capabilities demonstrate GFFM's potential to advance research and practical applications in agriculture, ecology, environmental monitoring, and urban development.

B. Robustness of Less Input Data

The GFFM algorithm employs multiple time series data as inputs to achieve better reconstruction results. It's evident that changes in the number of image pairs will affect the reconstruction results. By modifying the number of image pair inputs and predicting the remaining image pairs, we can perform a quantitative analysis of the result accuracy. This was compared with the ROBOT [52], HANTS [12], and GF-SG [40] techniques.

Fig. 11 illustrates the accuracy of reconstruction results in relation to different quantities of input data. Notably, the GFFM and ROBOT algorithms both demonstrate commendable stability and a gradual decrease in precision as the input data reduces, with GFFM surpassing ROBOT in terms of accuracy. In contrast, the HANTS method exhibits fluctuations

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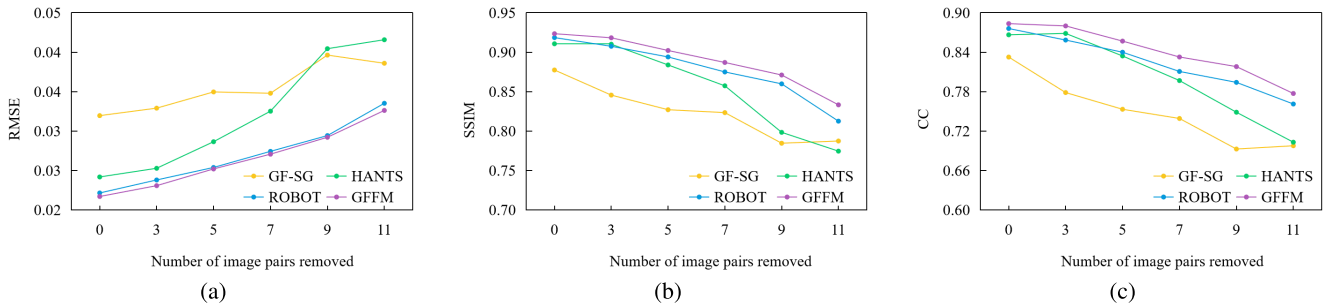


Fig. 11. Evaluation metrics of predicted results with different amounts of input data. Values were averaged over six bands. (a) RMSE value. (b) SSIM value. (c) CC value.

TABLE V
OVERALL COMPUTATION TIME OF 16 PREDICTED IMAGES FOR FOUR COMPARATIVE METHODS ON CIA SITE IN SECTION III-B

	GFFM	HANTS	VIPSTF	ESTARFM
Processing time	**41min**	54min	132min	117min
Programming language	Javascript(GEE)	Javascript(GEE)	Matlab(local)	IDL(local)
Memory usage (Uint8)	**48.3MB**	52MB	58.6MB	58.6MB

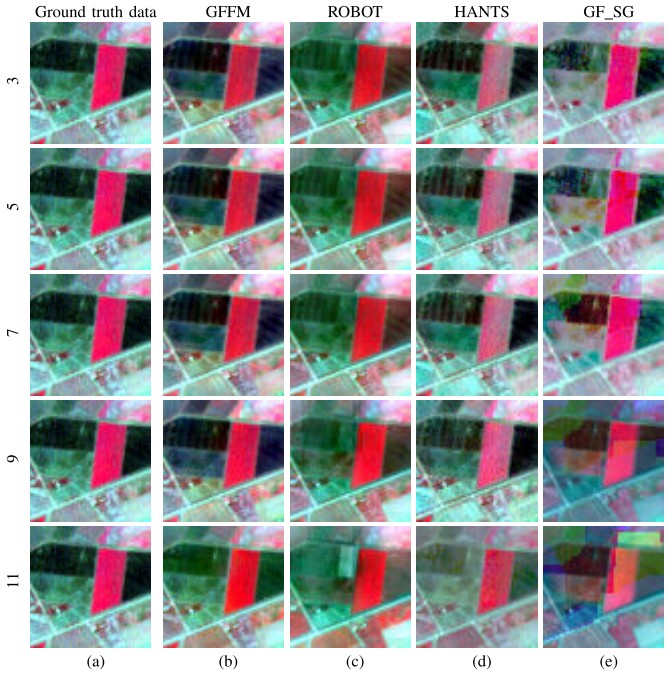


Fig. 12. Predicted results with different amounts of input data in subarea on CIA dataset. The reference Landsat data acquired on August 7, 2002 is (a). (b)–(e) GFFM, ROBOT, HANTS, and GF_SG predicted results, respectively.

and a faster decline as the input data decreases. The GF-SG algorithm, when applied to surface reflectance fusion, exhibited subpar performance and increased noise as the number of input images decreased. This is likely due to the fact that reconstruction algorithms built on phenological data are not well-suited for reflectance spectral images with weak correlations [45]. As shown in Fig. 12, the visualization results of different input data in the subarea of site 1 are displayed. It can be observed that the results of GFFM under different conditions are closest to the actual reference image.

C. Superiority of Daily-Scale Predictions

The method of synthesis at the daily scale possesses a high temporal resolution, enabling it to keenly detect minute changes on the surface of the Earth. Compared to this, the monthly scale synthesis method like HISTARFM might lead to the blurring of information at certain time points due to the consolidation of data from the entire month. In forecasting data at the monthly scale, we adopted two approaches: one is the direct use of synthesized data at the monthly scale, and the other involves feeding the original data into GFFM. The first approach involves directly utilizing the synthesized data at the monthly scale for predicting outcomes, whereas the second approach first acquires the results data on a daily basis, followed by synthesis at the monthly scale. In the prediction for a specific month, we take all the data excluding that from the month in question as input for forecasting the outcome. The computed results of the indicators are presented in Table VI, using the data for February and September from the CIA dataset as the data for prediction at the monthly scale. The results indicate that the data synthesized at the monthly scale through prediction at the daily scale outperforms the forecast data obtained directly through synthesis at the monthly scale in terms of predictive efficacy. This discovery suggests that the method of synthesis at the daily scale is capable of capturing the spatiotemporal changes on the Earth's surface in a more comprehensive manner, providing more precise input information for the outcomes, thereby rendering the data synthesized at the monthly scale more accurate. This further underscores the significance of the synthesis method at a daily scale in the fields of geoscience research and environmental monitoring.

D. Robustness of Cloud Contamination

To verify the robustness of GFFM to cloud contamination, an experiment was performed using the CIA dataset. The Landsat dataset acquired on April 1, 2002 shown in Fig. 13(a), is affected by thin cloud cover, and the black holes mask is used to process the data. The Landsat image acquired on April 10, 2002, is used as the prediction image, and the input data of the algorithm is the same as in Section III-B. For thin cloud pollution, the harmonic model and weight function in the GFFM method smooth the reflectance values, effectively reducing the impact of clouds and preserving the integrity of the underlying data. Fig. 13 presents the predictive outcomes of six algorithms, revealing that the method proposed in this article offers the most effective treatment for thin cloud processing, followed by the ROBOT algorithm. For the black holes mask processing section in the bottom right corner, it is

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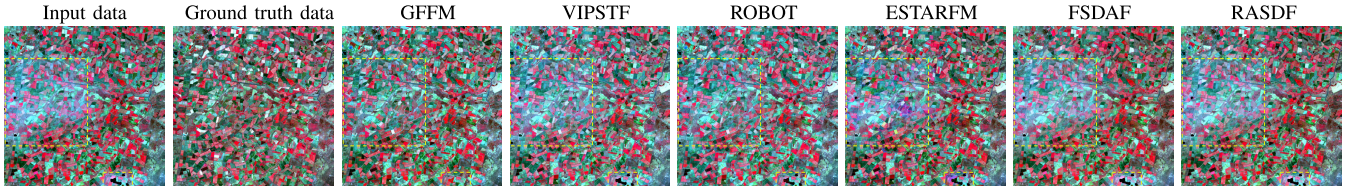


Fig. 13. Visual comparison of different predicted results on cloud contamination experiment. (a) The input Landsat data was acquired on April 1, 2002. (b) Ground truth Landsat data was acquired on April 10, 2002. (c)–(h) GFFM, VIPSTF, ROBOT, ESTARFM, FSDAF, and RASDF predicted results, respectively.

TABLE VI

MONTHLY PREDICTION OF GFFM IN FEBRUARY AND SEPTEMBER ON THE CIA DATASET. THE OPTIMAL VALUES ARE PRESENTED IN BOLD

	Monthly Prediction						Daily Prediction Composition					
	Month 2			Month 9			Month 2			Month 9		
Band	RMSE	SSIM	CC	RMSE	SSIM	CC	RMSE	SSIM	CC	RMSE	SSIM	CC
Ideal	0	1	1	0	1	1	0	1	1	0	1	1
blue	0.0119	0.9525	**0.9186**	0.0064	0.9704	0.9038	**0.0103**	**0.9569**	0.9171	**0.0062**	**0.9743**	**0.9178**
green	0.0209	0.9064	0.8603	0.0090	0.9525	0.8895	**0.0178**	**0.9153**	**0.8674**	**0.0085**	**0.9593**	**0.9051**
red	0.0350	0.8892	0.9003	0.0143	0.9378	0.9163	**0.0281**	**0.9103**	**0.9093**	**0.0122**	**0.9545**	**0.9387**
NIR	0.0497	0.8079	0.8176	**0.0248**	0.9269	0.9288	**0.0416**	**0.8900**	**0.8950**	0.0249	**0.9335**	**0.9350**
SWIR1	0.0355	**0.9431**	**0.9435**	0.0345	0.8925	0.8826	**0.0328**	0.9407	0.9407	**0.0340**	**0.9077**	**0.8998**
SWIR2	0.0268	**0.9511**	**0.9512**	0.0302	0.9095	0.9024	**0.0260**	0.9505	0.9498	**0.0277**	**0.9307**	**0.9243**

TABLE VII

ABLATION STUDY OF GFFM ON THE CIA DATASET. THE VALUES WERE AVERAGED WITH ALL PREDICTIONS OVER SIX BANDS AND OPTIMAL VALUES ARE PRESENTED IN BOLD

Band	GFFM			L_{fit}			MOD^{30}		
	RMSE	SSIM	CC	RMSE	SSIM	CC	RMSE	SSIM	CC
Ideal	0	1	1	0	1	1	0	1	1
blue	**0.0097**	**0.9522**	**0.8478**	0.0103	0.9464	0.8336	0.0120	0.9118	0.7125
green	**0.0115**	**0.9349**	**0.8457**	0.0131	0.9235	0.8232	0.0156	0.8685	0.6873
red	**0.0165**	**0.9222**	**0.8922**	0.0198	0.9066	0.8747	0.0258	0.8047	0.7262
NIR	**0.0278**	**0.9287**	**0.9265**	0.0319	0.9092	0.9016	0.0447	0.7776	0.7724
SWIR1	**0.0352**	**0.8872**	**0.8793**	0.0385	0.8715	0.8644	0.0460	0.8099	0.8015
SWIR2	**0.0295**	**0.9153**	**0.9089**	0.0316	0.9069	0.9011	0.0411	0.8302	0.8202

evident that only the GFFM method successfully eliminates interference to predict the results. Moreover, it is observed that the ESTARFM algorithm leads to some distortion in the results due to the disruption caused by cloud contamination. For missing data, the GFFM model operates similar to clear image inputs, but the reduced observation count for affected pixels alters the regression parameters and harmonic frequency selection. These adjustments ensure accurate reconstruction despite data gaps.

E. Ablation Studies

To assess the effectiveness of each step in the proposed method, an ablation experiment was performed on the CIA data in site 1, with the results outlined in Table VII. The results from Table VII show that compared to the individual temporal harmonic fitting (L_{fit}) and MODIS data correction results (MOD^{30}), the complete model using the BMA for weighted calculation significantly improves the accuracy of the reconstruction results, reflecting the effectiveness of fusion-filling strategy versus a single filling or fusion method.

F. Further Improvements

The method introduced in this article serves as a straightforward pixel-wise computational model, offering superior computational efficiency because it solely processes individual pixels without taking into account the neighborhood of pixels. The execution of this method on GEE further diminishes the processing time required for comprehensive image fusion tasks. However, it is understood that information from neighboring regions can offer more context, which is advantageous for identifying spatial patterns and dependencies in images [26], [52]. This can result in improved understanding and prediction of target pixels in fusion predictions. However, the architecture of GEE creates a challenge for partitioning the dataset into smaller units and utilizing a moving window for iterative calculations on each pixel. Thus, devising a better approach to this issue is a point of consideration for future advancements.

Furthermore, as a data-driven reconstruction method, GFFM relies on the availability of sufficient satellite observations, which can constrain its application in regions with minimal data or during short observation periods. Alternately, the GEE platform offers access to long-term surface reflectance datasets, including Landsat 4, 5, 7, and 8, as well as the harmonized Landsat-Sentinel-2 product [54]. To further enhance the model's applicability, integrating synthesis techniques [44], [55] could address data gaps and improve performance in areas with limited observations. These future developments could expand the robustness of GFFM, making it more effective in challenging data-sparse environments.

V. CONCLUSION

The research introduces a fusion-filling method that can be easily implemented in GEE. It leverages the initial fusion outcomes of Landsat and MODIS data, integrated with Landsat temporal fitting data. The ultimate high-quality reconstruction results are derived through an ingenious weighting function. GFFM effectively capitalizes on cloud computing platforms to significantly reduce the steps required in the local preprocessing stage of large amounts of data. Moreover, it offers a

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practical advantage in long-term calculations. GFFM has been benchmarked against multiple cutting-edge reconstruction algorithms on two typical datasets, and it has demonstrated superior accuracy, thereby confirming the fusion proficiency of GFFM. Moreover, the ablation experiments carried out on various elements of the method design suggest that each component of the GFFM has an impact on the reconstruction results. The comparative experiments with HISTARFM and the experiments involving cloud-contaminated image inputs underscore the effectiveness of the method presented in this article in utilizing images with partial cloud contamination. Therefore, there is a potential to produce high-resolution long-term seamless surface reflectance data for further applications such as biomass estimation and dynamic land cover monitoring.

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